01/07/2018
SUNDAY
19.30 hrs to 20.30 hrs.

Statistical Methods for Engineers

Unit - II - Testing hypothesis

* Student - t - Test — (Mean)
* F - Test — (Variance)
* χ^2 - Test — &
* Test based on Normal (Mean & Proportion)
* ~~Analysis of Variance~~
  * ~~One Way~~

Testing Hypothesis.

* Population
* Sample

Test of Hypothesis

* Null Hypothesis (H_0) ($\theta = \theta_0$)

(There is no significant difference between two sample statistics).

* Alternate Hypothesis (H_1) ($\theta \neq \theta_0$) ($\theta > \theta_0$) ($\theta < \theta_0$)

(There is a significant difference between two sample statistics)

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Critical Region (Region of Rejection)

$\alpha = 1\%$ (or) 2% (or) 5% (or) 10% is called as Level of significance (LOS)

Error in Hypothesis testing

Type - I error
Type - II error

Types of tailed test

* one tailed

Right tailed   left tailed

(Common) $H_0 : \theta = \theta_0$ (Null Hypothesis)

i) $H_1 : \theta \neq \theta_0$ (Alternate Hypothesis) - Two tailed test

ii) $H_1 : \theta > \theta_0$ - Right tailed test

iii) $H_1 : \theta < \theta_0$ - left tailed test

Procedure for testing Hypothesis

1. Null Hypothesis H_0 is defined

2. Alternate Hypothesis H_1 is defined

3. LOS $= \alpha$ % & Table value Z_α is fixed

4. Test Statistic

$$\text{Calculated value} = Z = \frac{t - E(t)}{S.E(t)}$$

$E(t)$ — Mean
$S.E(t)$ = Standard deviation
t — statistic

5. Compare $|Z|$ with table value (Z_α)

6. Conclusion

i) If $|Z| < Z_\alpha$
then H_0 is accepted.
i.e., H_1 is rejected

ii) If $|Z| > Z_\alpha$
then H_0 is rejected
i.e., H_1 is accepted.

Types of Sample

① Large Sample ($n \geq 30$)

② Small Sample $n \leq 30$

Unit - II

Outline

Large sample

* Z - test (6 types)

Small Sample

* Student - t - test (4 types)
* F test.

χ^2 test

* Goodness of fit
* Independence of Attributes

Large Sample ($n \geq 30$)

Z - Test

① Between Sample proportion (p) & Population proportion (P)

Test Statistic

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

n — Size of the Sample
p — Sample proportion
P — Population proportion

($\because P+Q = 1$
$Q = 1-P$)

② Between two sample proportion (p_1 & p_2)

$$Z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$p_1 \rightarrow$ 1st sample proportion
$p_2 \rightarrow$ 2nd Sample Proportion

$P \rightarrow$ Population Proportion

$$= \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \quad Q = 1-P$$

n_1 — 1st Sample size, $n_2 \rightarrow$ 2nd Sample Size.

Test Statistic

③ Between Sample ($\bar{x}$) Mean & population mean (μ)

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Sample mean — $\bar{X}$

Population Mean — μ

σ — S.D (Standard Deviation)

n — Sample size.

④ Between two sample mean ($\bar{X}_1$ & $\bar{X}_2$)

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$\bar{X}_1$ — Mean of sample-I

$\bar{X}_2$ — Mean of sample-II

σ_1 — S.D of sample-I

σ_2 — SD of sample-II

n_1 — Size of sample-I

n_2 — Size of ~~sample~~-II

(5)	Between Sample S.D & Population S.D (σ)	$$Z = \frac{S - \sigma}{\sigma / \sqrt{2n}}$$ S – Sample S.D σ – Population S.D n – size of the sample.
(6)	Between two Sample SD's. (or) Between S.D's of two Samples	$$Z = \frac{S_1 - S_2}{\sigma\sqrt{\frac{1}{2n_1} + \frac{1}{2n_2}}}$$ where σ is Known S_1 & S_2 – S.D's of Sample I & II respectively n_1 & n_2 → Size of the Samples I & II respectively

Note → If σ is not known

Put $\sigma = \sqrt{\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}}$

ie,).

$$Z = \frac{S_1 - S_2}{\sqrt{\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}}\sqrt{\frac{1}{2n_1} + \frac{1}{2n_2}}}$$

(or)

$$= \frac{S_1 - S_2}{\sqrt{\frac{S_1^2}{2n_2} + \frac{S_2^2}{2n_1}}}$$

Note

For right (or) (Left) we have to follow smallest side.

Problems

① The fatality rate of typhoid patients is believed to be 17.26 percent. In a certain year 640 patients suffering from typhoid were treated in a metropolitan Hospital & only 63 patients died. Can you consider the Hospital is efficient?

Sol

Step: 1.

Null Hypothesis is

$H_0 : p = P$

Step-2

$H_1 : \cancel{p \neq P}\ p < P$

left tailed test to be followed

Step: 3

At $\alpha = 1\%$ LOS

$Z_\alpha = -2.33$

table value $= |Z_\alpha| = 2.33$ at $\alpha = 1\%$ LOS

Step: 4

Test Statistic

$$Z = \frac{p - P}{\sqrt{PQ/n}}$$

Given

$$p = \frac{63}{640} = 0.0984$$

$$P = 17.26\% = \frac{17.26}{100} = 0.1726$$

$$Q = 1 - 0.1726 = 0.8274 \qquad n = 640$$

$$Z = \frac{0.0984 - 0.1726}{\sqrt{\frac{0.1726 \times 0.8274}{640}}}$$

$$= -4.96$$

$|Z| = 4.96$ = Calculated value

Step, 5

$$|z| > |z_\alpha|$$

Calculated value is greater than the table value.

Conclusion

$\therefore$ H_0 is rejected & H_1 is accepted.

~~Conclusion~~

$\therefore$ The Hospital is efficient.

practice

(12) A manufacturer of light bulbs claims that on the averge 2 % of the bulbs manufactured by his firm are defective. A random sample of 400 bulbs contained 13 defective bulbs. On the basis of the this sample, can you support the manufacturer's claim at 5 % level of significance?

Solution:

Step: 1

Null Hypothesis

$H_0 : p = P$

Step: 2

$H_1 : p < P$ left tailed test to be followed

Step: 3

At $\alpha = 5\%$ LOS

$Z_\alpha = -1.64$

Table Value $\} = |Z| = 1.64$ at $\alpha = 5\%$ LOS

Step: 4

Test Statistic

$$Z = \frac{p - P}{\sqrt{PQ/n}}$$

Given

$$p = \frac{13}{400} = 0.0325$$

$$P = 2\% = \frac{2}{100} = 0.02$$

$P + Q = 1$

$Q = 1 - P = 1 - 0.02$ | $n = 400$

$Q = 0.98$

$$Z = \frac{0.0325 - 0.02}{\sqrt{\frac{0.02 \times 0.98}{400}}} = \frac{0.0125}{\sqrt{\frac{0.0196}{400}}}$$

$$= \frac{0.0125}{\sqrt{0.000049}} = \frac{0.0125}{0.007}$$

$$Z = 1.785$$

$|Z| = 1.785$ = Calculated value.

Step: 5 ~~|Z| < |Z|~~ $|Z| > |Z_\alpha|$

Calculated value is greater than the table value.

Step: 6

Conclusion:

$\therefore$ H_0 is rejected & H_1 is accepted

$\therefore$ We can support the manufacturer.

07.07.18
Saturday (3)
4.30PM
to 5.30PM

In a large city A, 20 percent of a random sample of 900 school boys had a slight physical defect. In another large city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant?

Sol

Given,

$$p_1 = \frac{20}{100} = 0.2 \ \& \ p_2 = 0.185$$

$$n_1 = 900, \quad n_2 = 1600$$

Step: 1 $H_0: P_1 = P_2$

Two tailed test to be followed

$H_1: P_1 \neq P_2$

Step: 2

$LOS = 5\% = \alpha$

Table value $= Z_\alpha = 1.96$ where $\alpha = 5\%$.

$|Z_\alpha| = 1.96$

Step: 3

Test Statistic

$$Z = \frac{P_1 - P_2}{\sqrt{PQ\left(\frac{1}{n_1} + 1/n_2\right)}}$$

$$P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

$$= \frac{180 + 296}{900 + 1600} = 0.1904$$

$$Q = 1 - 0.1904 = 0.8096$$

$$Z = \frac{0.2 - 0.185}{\sqrt{(0.1904)(0.8096)\left(\frac{1}{900} + \frac{1}{1600}\right)}}$$

$$Z = 0.92$$

$$|Z| = 0.92$$

Step: 4

$$|Z| < |Z_\alpha|$$

$\therefore$ Ho is accepted

$\therefore$ There is no significant difference.

(4) A Sample of 100 Students is taken from a large population. The mean height of the Students in this Sample is 160 cm. Can it be reasonably regarded that, in the population, the mean height is 165 cm & S.D. is 10 cm?

Sol

Given, $\bar{x} = 160$, $n = 100$

$\mu = 165$, $\sigma = 10$

Step: 1

$H_0 : \bar{x} = \mu$

$H_1 : \bar{x} \neq \mu$

(Two tailed test to be followed)

Step: 2

$LOS = \alpha = 5\%$

Table value $= Z_{\alpha} = 1.96$

$|Z_{\alpha}| = 1.96$

Step: 3

Test Statistic $= Z = \dfrac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

$$= \frac{160-165}{10/\sqrt{100}} = \frac{-5}{10/10}$$

$$z = -5$$

$$|z| = 5$$

Step: 4

$$|z| > |z_\alpha|$$

$\therefore$ H_0 is rejected

& H_1 is accepted

Given data is not considered reasonable.

8/07/18

(1) A simple sample of heights of 6400 English men has a mean of 170 cm and a SD of 64 cm while a simple sample of heights of 1600 American has a mean of 172 cm & a SD of 6.3 cm. Do the data indicate that Americans are, on the average taller than the English men?

Sol

Given $n_1 = 6400$, $n_2 = 1600$

$\bar{x}_1 = 170$ cm $\bar{x}_2 = 172$ cm

$S_1 = 6.4$ cm $S_2 = 6.3$ cm

Step: 1

$H_0: \bar{x}_1 = \bar{x}_2$

$H_1: \bar{x}_1 < \bar{x}_2$

Left tailed test to be followed.

Step: 2

$LOS = 5\% = \alpha$

$Z_\alpha =$ table value $= -1.645$

$|Z_\alpha| = 1.645$

Step: 3

Test Statistic

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$= \frac{170 - 172}{\sqrt{\frac{(6.4)^2}{6400} + \frac{(6.3)^2}{1600}}}$$

$= -11.32$ & $|Z| = 11.32$

Step: 4

$|Z| > |Z_\alpha|$

H_0 is rejected & H_1 is accepted

$\therefore$ Average height of americans is taller than the average height of Englishmen.

(2) A manufacturer of electric bulbs, according to a certain process, finds the SD of the life of lamps to be 100 hours. He wants to change the process, If the new process results in a smaller variation in the life of lamps, In adopting a new process, a sample of 150 bulbs gave an S.D of 95 hrs. Is the manufacturer justified in changing the process?

Sol

$\sigma = 100$ = Population S.D

$n = 150$

$s = 95$ hrs = Sample S.D

Step:1

$H_0 : \sigma = s$ $\quad s = \sigma$

$H_1 : \cancel{\sigma > s}$ $\quad s < \sigma$

Left ~~Right~~ tailed test to be followed

Step: 2

LOS = 5% = α

table value = $Z_\alpha = -1.645$

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$$|Z_\alpha| = 1.645$$

Step:3

Test Statistic

$$Z = \frac{S - \sigma}{\sigma / \sqrt{2n}}$$

$$= \frac{95 - 100}{100 / \sqrt{300}}$$

$$= -0.866$$

$$|Z| = 0.866$$

Step:4

$$|Z| < |Z_\alpha|$$

$\therefore$ Ho is accepted

Conclusion

The manufactures is not justified in changing the Process.

③ The SD of a random sample 1000 is found to be 2.6 & the SD of another random sample of 500 is 2.7. Assuming the samples to be independent. Find whether the (two sample could have come from populations with the same SD.)

Sol

Given,

$n_1 = 1000$ $\qquad$ $n_2 = 500$

$s_1 = 2.6$ $\qquad$ $s_2 = 2.7$

Step: 1

$H_0 : S_1 = S_2$

$H_1 : S_1 \neq S_2$

Two tailed test to be followed.

Step: 2

LOS : $\alpha = 5\%$

Table value $= Z_\alpha = 1.96$

$|Z_\alpha| = 1.96$

Step: 3

Test Statistic

$$Z = \frac{S_1 - S_2}{\sqrt{\frac{S_1^2}{2n_2} + \frac{S_2^2}{2n_1}}}$$ when σ is not known

$$= \frac{2.6 - 2.7}{\sqrt{\frac{(2.6)^2}{1000} + \frac{(2.7)^2}{2000}}} = -0.98$$

$|Z| = 0.98$

Step: 4

$|Z| < |Z_\alpha|$

$\therefore$ H_0 is accepted

$\therefore$ Two Sample could have Came from Populations with same S.D.

Test of significance for small Samples $(n < 30)$

Student's t - Distribution

Pdf

$$f(t) = \frac{1}{\sqrt{v}\, B\left(\frac{v}{2}, \frac{1}{2}\right)} \left(1 + \frac{t^2}{v}\right)^{-(v+1)/2}, \quad -\infty < t < \infty$$

Gamma function

v — Degrees of freedom.

Note

* 1st order mean = Dof = $n-1$ for size $= n$
* 2nd order mean (SD) = Dof = $n-2$ for size $= n$

Properties of t - distribution

* Probability curve is similar to standard normal curve.
* As $n \to \infty$, t distribution tends to standard normal distribution

3. Mean of the t - distribution is zero.

4. Variance of the t-distribution is $\frac{v}{v-2}$, if $n > 2$ & is greater than 1, but it tends to 1 as $v \to \infty$.

Uses of t - distribution

The t - distribution is used to test the significance of the difference between

i) the mean of small sample & the mean of the population

ii) the means of two small samples

iii) The coefficient of correlation in the small sample & that in the population, assumed zero.

S.D

$$s^2 = \frac{1}{n} \Sigma (x_i - \bar{x})^2 \quad (or) \quad \frac{1}{n} \Sigma x_i^2 - \left(\frac{1}{n} \Sigma x_i\right)^2$$

Students t-test

① Sample mean $\bar{x}$ & population mean μ

$$t = Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

DOF = V = n-1

(or) $$Z = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

② Difference between means of two small samples

$$t = Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{1/n_1 + 1/n_2}}$$

where σ is known

DOF $= n_1 - 1 + n_2 - 1$
$= n_1 + n_2 - 2$

If σ is not known but S_1 & S_2 are known

then $$\sigma = \sqrt{\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}}$$

i)

If $n_1 = n_2 = n$ & the samples are independent (samples are not at all related)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2 + S_2^2}{n-1}}}$$

DOF = 2n-2

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ii) If $n_1 = n_2 = n$ & if the pairs of values of x_1 & x_2 are related (or) dependent

$H_0 : \bar{d} = 0$ $\cancel{(\bar{x} - \bar{y} = 0)}$

where $\bar{d} = \bar{x} - \bar{y}$

<u>Test Statistic</u>

$$t = \frac{\bar{d}}{S/\sqrt{n-1}} \quad \& \text{ DoF} = n-1$$

where $d_i = x_i - y_i$ for $i = 1, 2 \ldots$

$\bar{d} = \bar{x} - \bar{y}$ &

$$S = \frac{1}{n} \sum_{i=1}^{n} \left(d_i - \bar{d}\right)^2$$

$$(\text{or}) \quad \frac{1}{n} \sum_{i=1}^{n} d_i^2 - \left(\frac{1}{n} \sum_{i=1}^{n} d_i\right)^2$$

14/07/18
SATURDAY
4.00PM.

Problems - t - Test

(1) A certain injection administered to each of 12 patients in the following increases of blood pressure:

5, 2, 8, −1, 3, 0, 6, −2, 1, 5, 0, 4

Can it be concluded that the injection will be, in general accompanied by an increase in BP? ~~with mean 2.58~~.

Sol

Step:1

$$\text{Mean} = \bar{x} = \left(\frac{\Sigma x}{n}\right) = \frac{31}{12} = 2.58$$

$$S.D = \sqrt{\frac{1}{n}\Sigma x^2 - \left(\frac{1}{n}\Sigma x\right)^2}$$

x	x^2
5	25
2	4
8	64
−1	1
3	9
0	0
6	36
−2	4
1	1
5	25
0	0
4	16

$\Sigma x = 31$, $\Sigma x^2 = 185$

$$S.D = \sqrt{\frac{1}{12}(185) - (2.58)^2}$$

$$S = 8.76$$

~~St~~ $H_0 : \bar{x} = \mu$ $(\mu = 0)$

$H_1 : \bar{x} > \mu$

Step:2

LOS = 5% = α is for one tailed but for two tailed $\alpha = 10\%$

t_α with $V = n-1 = 11$ (DoF)

$t_{0.10}$ with DoF 11 = 1.80

Step:3

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n-1}}$$

$$= \frac{2.58 - 0}{8.76/\sqrt{11}}$$

$|t| = 2.89$

Step:4

$|t| > t_\alpha$

$\therefore$ H_0 is rejected $\therefore$ The injection is accompanied by an increase in BP.

H_1 is accepted

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(2) Two independent Sample of Size 8 & 7 Contained the following Values

Sample-I : 19 17 15 21 16 18 16 14

Sample-II : 15 14 15 19 15 18 16

Is the difference between the Sample means significant?

Sol

To find: $\bar{x}_1, \bar{x}_2, s_1, s_2$

S-I

$n_1 = 8$
$n_2 = 7$

x_1	x_1^2	x_2	x_2^2
19	361	15	225
17	289	14	196
15	225	15	225
21	441	19	361
16	256	15	225
18	324	18	324
16	256	16	256
14	196		
$\Sigma x_1 = 136$	$\Sigma x_1^2 = 2348$	$\Sigma x_2 = 112$	$\Sigma x_2^2 = 1812$

s_1 = SD of Sample-I

$$= \sqrt{\frac{1}{n_1}\Sigma x_1^2 - \left(\frac{1}{n_1}\Sigma x_1\right)^2}$$

$$= \sqrt{\frac{2348}{8} - \left(\frac{136}{8}\right)^2}$$

$$= \sqrt{293.5 - (289)}$$

$$= \sqrt{4.5}$$

$$= 2.1213$$

$$\bar{x}_1 = \frac{136}{8} = \frac{\Sigma x_1}{n_1} = 17$$

S.D of IInd Sample $= S_2 = \sqrt{\frac{1}{n_2}\Sigma x_2^2 - \left(\frac{1}{n_2}\Sigma x_2\right)^2}$

$= \sqrt{\frac{1812}{7} - \left(\frac{112}{7}\right)^2}$

$= \sqrt{258.857 - 256}$

$S_2 = \sqrt{2.857} = 1.690$

$\bar{x}_2 = \frac{\Sigma x_2}{n} = \frac{112}{7} = 16$

Step: 2

$H_0: \bar{x}_1 = \bar{x}_2$

$H_1: \bar{x}_1 \neq \bar{x}_2$

Two tailed test to be used.

Step: 3

$LOS = \alpha = 5\%$

t_α with $DOF = v = n_1 + n_2 - 2 = 8 + 7 - 2 = 13$

$t_{0.05}$ with $DOF = 13 = 2.16$ (table value)

Step: 4 (σ is not known)

t-Test $= t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

$$= \frac{17 - 16}{\sqrt{\left(\frac{8(4.5)+7(2.857)}{13}\right)\left(\frac{1}{8}+\frac{1}{7}\right)}}$$

$$= \frac{1}{\sqrt{(4.3076)(0.2678)}}$$

$$= \frac{1}{\sqrt{1.1538}} = \frac{1}{1.0741}$$

$$|t| = 0.931$$

Step:5

$|t| < t_\alpha$

$\therefore$ H_0 is accepted

H_1 is rejected.

$\therefore$ There is no significant difference between two sample mean.

21/07/2018
SATURDAY
4:00 P.M.

v_1

Snedecor's F - Distribution :-

A r.v $F \sim$ F-distribution if its pdf

$$f(F) = \frac{\left(\frac{v_1}{v_2}\right)^{v_1/2} \cdot F^{\frac{v_1}{2}-1}}{B\left(\frac{v_1}{2}, \frac{v_2}{2}\right)\left(1+\frac{v_1 F}{v_2}\right)^{\frac{v_1+v_2}{2}}}$$

where $F > 0$

Properties

* Square of t variate with n Dof $\sim$ F distribution with 1 to n Dof

* Mean $= \dfrac{V_2}{V_2 - 2}$ where $V_2 > 2$

* Variance $= \dfrac{2V_2^2 (V_1 + V_2 - 2)}{V_1 (V_2 - 2)^2 (V_2 - 4)}$

Where $V_2 > 4$

Assumption

σ_1^2 — variance of population I

σ_2^2 — variance of " II

S_1^2 — variance of sample I

S_2^2 — variance of " II

n_1 — Size of the sample - I

n_2 — Size of the sample - II

Note - 1

σ_1^2 & σ_2^2 are not known but

S_1^2 & S_2^2 are known

$$\sigma_1^2 = \frac{n_1 S_1^2}{n_1 - 1}$$ with Dof of $v_1 = n_1 - 1$

$$\sigma_2^2 = \frac{n_2 S^2}{n_2 - 1}$$ with Dof of $V_2 = n_2 - 1$

Note - 2

* $$F \text{ ratio} = \frac{\sigma_1^2}{\sigma_2^2}$$ when $\sigma_1^2 > \sigma_2^2$

* Alway $F > 1$

* Dof $= (V_1, V_2)$

where $V_1 = n_1 - 1$

$V_2 = n_2 - 1$

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Sums

(1) Two independent samples of eight & seven items respectively had the following values of the variable

Sample-I	9	11	13	11	15	9	12	14
Sample-II	10	12	10	14	9	8	10	

Do the two estimates of population variance differ significantly at 5% LOS.

Sol

$$s_1^2 = var(s_1) = \frac{1}{n_1}\sum x_1^2 - \left(\frac{1}{n_1}\sum x_1\right)^2$$

$$|||^{ly}\ s_2^2 = var(s_2) = \frac{1}{n_2}\sum x_2^2 - \left(\frac{1}{n_2}\sum x_2\right)^2$$

Given $n_1 = 8$, $n_2 = 7$

	x_1	x_2	x_1^2	x_2^2
	9	10	81	100
	11	12	121	144
	13	10	169	100
	11	14	121	196
	15	9	225	81
	9	8	81	64
	12	10	144	100
	14		196	
Σ	94	73	1138	785

$$S_1^2 = \frac{1}{n_1}\Sigma x_1^2 - \left(\frac{1}{n_1}\Sigma x_1\right)^2$$

$$= \frac{1138}{8} - \left(\frac{94}{8}\right)^2$$

$S_1^2 = 4.19$ = Variance of S-I

$$S_2^2 = \frac{1}{n_2}\Sigma x_2^2 - \left(\frac{1}{n_2}\Sigma x_2\right)^2$$

$$= \frac{785}{7} - \left(\frac{1}{7}73\right)^2$$

$$= 3.39$$

$$\sigma_1^2 = \frac{n_1 S_1^2}{n_1 - 1} = \frac{8(4.19)}{7}$$

$$= 4.79$$

$$\sigma_2^2 = \frac{n_2 S_2^2}{n_2 - 1} = \frac{7(3.39)}{6} = 3.96$$

Step: 2

$H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

Step: 3

Table value of F at 5% LOS

$F_{\alpha = 0.005}$ with DoF $(\nu_1 = 7, \nu_2 = 6)$ $= 4.21$

Step: 4

$$F = \frac{\sigma_1^2}{\sigma_2^2} = \frac{4.79}{3.96} = 1.21$$

$\therefore$ Since $F_{cal} < F_{table}$

$\therefore$ H_0 is accepted

$\therefore$ There is no significan between V_1^2 & V_2^2.

Chisquare Distribution

Pdf :
$$f(\chi^2) = \frac{1}{2^{\nu/2}\sqrt{\nu/2}} (\chi^2)^{\nu/2 - 1} e^{-\chi^2/2}$$

with mean $= \nu$

Variance $= 2\nu$

Uses of χ^2-test

1. To Test ~~a~~ the goodness of fit
2. Independence of attributes.

Goodness of fit

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

$O_i \rightarrow$ observed frequency

$E_i \rightarrow$ Expected frequency

(1) The table given below states that the no. of air craft accidents ~~that~~ occured during the various days of a week. Test whether the accidents are uniformly distributed over the week.

Table

Day	Mon	Tues	Wed	Thu	Fri	Sat
No. of accidents	15	19	13	12	16	15

Sol

H_0 : Accidents occurs uniformly over the week.

H_1 : Not occurs uniformly

$$E_i = \frac{\sum x_i}{n} = \frac{90}{6} = 15$$

O_i	15	19	13	12	16	15
E_i	15	15	15	15	15	15

$$\psi^2 = \sum_{i=1}^{6} \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{0^2}{15} + \frac{4^2}{15} + \frac{2^2}{15} + \frac{(-3)^2}{15}$$

$$+ 1^2/15 + 0^2/15$$

$= \frac{1}{15}(16+4+9+1) = 2$

$DoF = \nu = 6-1 = 5$
$(n-1)$

$\therefore \chi^2$ (Calculated value) $= 12$

Table value: $\chi^2_{at\ 5\%\ LoS}$ with DoF $\nu = 5$

$= 11.07$

$\therefore \chi^2 < \chi^2_{5\%}$

$\therefore H_0$ is accepted

22/07/18

① Theory predicts that the proportion of beans in four groups A, B, C, D should be 9:3:3:1. In an experiment among 1600 beans the no's in the four groups were 882, 313, 287 & 118. Dog the experiment support the theory.

Sol: Step-1

H_0: The experiment supports the theory

Step-2

O_i	E_i	$O_i - E_i$
882	$\frac{9}{16} \times 1600 = 900$	-18
313	$\frac{3}{16} \times 1600 = 300$	13
287	$\frac{3}{16} \times 1600 = 300$	-13
118	$\frac{1}{16} \times 1600 = 100$	18

$$\psi^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{18^2}{900} + \frac{13^2}{300} + \frac{13^2}{300} + \frac{18^2}{100}$$

$$= 4.73$$

Since $\sum E_i = \sum O_i$

$v = \text{DoF} = n-1 = 4-1 = 3$

Step-3

Table value $= \psi^2_{5\%}$ LoS with $v=3$

$= 7.82$

$\psi^2 < \psi^2_{5\%}$

H_0 is accepted.

(2) A survey of 320 families with five children each revealed the following distibution

No. of boys:	0	1	2	3	4	5
No. of girls:	5	4	3	2	1	0
No. of families	12	40	88	110	56	14

Is the result consistent with the hypothesis that male & female births are equally probable?

Sol

H_0: Male & female birth are equally probable.

$p = 1/2 = P(\text{male birth})$

$q = 1-p = 1/2$

$n = 5$ children

$P(X=x) = nC_x\, p^x q^{n-x}$

$P(X=r) = 5C_r (1/2)^r (1/2)^{5-r}$

$= 5C_r (1/2)^5$

Expected no. of families having r male children $= 320 \times P(x=r)$

$= 320 \times 5C_r (1/2)^5$

$= \cancel{320}^{10} \times 5C_r \times \frac{1}{\cancel{2\times2\times2\times2\times2}}$

$= 10 (5C_r)$

$\boxed{\therefore 5C_2 = \frac{5\times\cancel{4}^2}{\cancel{2}\times1} = 10}$

O_i	E_i	$O_i - E_i$
12	$10\times5C_0 = 10$	2
40	$10\times5C_1 = 50$	-10
~~88~~ 88	$10\times5C_2 = 100$	-12
110	$10\times5C_3 = 100$	10
56	$10\times5C_4 = 50$	6
14	$10\times5C_5 = 10$	4

$$\chi^2 = \sum_{i=1}^{6} \frac{(O_i - E_i)^2}{E_i} = \frac{4}{10} + \frac{100}{50} + \frac{144}{100} + \frac{100}{100} + \frac{36}{50} + \frac{16}{10}$$

$= 7.16$

Table value $= \chi^2_{5\%\ Los}$ with DoF $(6-1=5)$

$= 11.07$

$\therefore \chi^2 < \chi^2_{0.05}$

$\therefore H_0$ is accepted.

(3) Twelve dice ~~were~~ were thrown 4096 times & a throw of six was considered ~~as~~ ~~even~~ success. The observed frequencies were as given below.

No. of Successes	0	1	2	3	4	5	6	7 & above
Frequency	447	1145	1180	796	380	115	25	8

Test whether the dice were unbiased.

Sol

Step: 1 H_0: The dice were unbiased

Step: 2 $n(S) = 6$ In throwing a die where $S = \{1,2,3,4,5,6\}$

$P(\text{getting six}) = p = \frac{1}{6}$

$q = 1-p = 1 - 1/6 = 5/6$

$n =$ no. of dice $= 12$

$P(X=x) = nC_x \, p^x q^{n-x}$, $x = 0,1,2 \ldots 12$

$$P(X=x) = {}^{12}C_x \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{12-x}$$

$$= {}^{12}C_x \frac{1}{6^x} \cdot \frac{5^{12-x}}{6^{12-x}}$$

$$= {}^{12}C_x \cdot \frac{5^{12-x}}{6^{12}}$$

Expected no. of times in which r success are obtained

$$f = 4096 \times P(X=r)$$

$$= 4096 \times {}^{12}C_r \frac{5^{12-r}}{6^{12}}$$

$r = 0, 1, 2 \ldots 12$

O_i	E_i		Rounded E_i
447	$4096 \times {}^{12}C_0 \frac{5^{12}}{6^{12}}$	$= 459.39$	459
1145	$4096 \times {}^{12}C_1 \frac{5^{11}}{6^{12}}$	$= 1102.54$	1103
1180		1212.80	1213
796		808.53	809
380		363.84	364
115		116.43	116
25		27.17	27
8		5.3	5

$$\chi^2 = \sum_{i=1}^{8} \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{12^2}{459} + \frac{42^2}{1103} + \frac{33^2}{1213} + \frac{13^2}{889} + \frac{16^2}{364} + \frac{1^2}{116} + \frac{1^2}{32}$$

$$= 3.76$$

$DoF = v = n-1 = 8-1 = 7$

$\chi^2_{0.05}$ with $DoF = 6$ $= 14.067$

$\therefore \chi^2 < \chi^2_{0.05}$

$\therefore$ H_0 is accepted.

Type - II

χ^2 - test of Independence of Attributes

* $$\chi^2 = \sum_{i=1}^{m} \sum_{j=1}^{n} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$ Refer Table

Where $$E_{ij} = \frac{O_{i*} \times O_{*j}}{N}$$

* $$DOF = \nu = (m-1)(n-1)$$

① A total no. of 3759 individuals were interviewed in a public opinion Survey on a political proposal. Of them 1872 were men & the rest women. 2257 individual were infavour of the proposal & 917 were opposed to it. 243 men were undecided & 442 women were opposed to the proposal. Do you justify (or) contradict the hypothesis that there is no association between Sex & attitude?

	Favoured	opposed	undecided	Total
Men	1154	475	243	1872
women	1103	442	342	1887
	2257	917	585	3759

N

Sol

H_0: Sex & attitude are independent

O	E	$(O-E)^2/E$
1154	$\frac{1872 \times 2257}{3759} \simeq 1124$	$302/1124 = 0.80$
475	$\frac{1872 \times 917}{3759} \simeq 457$	$182/457 = 0.71$
243	$\frac{1872 \times 585}{3759} \simeq 291$	$482/291 = 7.92$
1103	$\frac{1887 \times 2257}{3759} \simeq 1133$	$302/1133 = 0.79$
442	$\frac{1887 \times 917}{3759} \simeq 460$	$182/460 = 0.70$
342	$\frac{1887 \times 585}{3759} \simeq 294$	$482/294 = 7.84$

$\chi^2 = 18.76$

for DoF row should consider

$V = (2-1)(\cancel{4}-1) = 1 \times 2 = 2$

$\chi^2_{0.05}$ with $(V = 2) = 5.99$

$\chi^2 > \chi^2_{0.05}$

$\therefore$ H_0 is rejected.

T-test For dependent data

(1) The following data relate to the marks obtained by 11 students in two test, one held at the begining of a year & the other at the end of the year after intensive coaching. Do the data indicate that the students have benefitted by coaching?

Test-1	19	23	16	24	17	18	20	18	21	19	20
Test-2	17	24	20	24	20	22	20	20	18	22	19

~~Then~~

Sol

Step-1 Let: x_1, x_2 — Marks in two tests.

$$d = x_1 - x_2$$

$\therefore$ The values of d are

d: 2, −1, −4, 0, −3, −4, 0, −2, 3, −3, 1
d^2: 4 1 16 0 9 16 0 4 9 9 1

$$\bar{d} = \frac{\Sigma d}{n} = \frac{-11}{11} = -1$$

$$s^2 = \frac{1}{n}\Sigma d^2 - \left(\frac{1}{n}\Sigma d\right)^2$$

$$= \frac{1}{n}\Sigma d^2 - \left(\bar{d}\right)^2 = \frac{69}{11} - (-1)^2$$

$$s^2 = 5.27$$

$$s = \sqrt{5.27} = 2.296$$

Step-2
$H_0: \bar{d} = 0 \quad (\bar{x}_1 = \bar{x}_2)$

$H_1: \bar{d} < 0 \quad (\bar{x}_1 < \bar{x}_2)$ one failed

Step-3
Table value $= t$ at 10% LoS
with (DoF $= v = 10$)
$= 1.81$

$$t = \frac{\bar{d}}{s/\sqrt{n-1}}$$

$$= \frac{-1}{2.296/\sqrt{10}} = -1.38$$

$$|t| = 1.38$$

$$|t| < t_{\alpha}$$

$\therefore$ H_0 is accepted

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